



Hurricane vulnerability modeling: Development and future trends

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ARTICLE INFO

Article history:

Received 5 September 2011

Received in revised form

30 November 2012

Accepted 7 December 2012

Available online 10 February 2013

Keywords:

Vulnerability

Catastrophe modeling

Uncertainty

Loss models

ABSTRACT

Catastrophe models help to evaluate the vulnerability of the building stock exposed to a hazard. This paper presents a history of the hurricane risk models in Florida, and discusses their relationship to the building codes. The first models were econometric, and failed to predict the insured building losses produced by hurricane Andrew. This led to a change in the loss projection paradigm and to the advent of modern catastrophe modeling. Advantages and challenges of the current methodologies are discussed, including the quality of input, validation, uncertainty, and scope of the outputs. The paper concludes with a brief overview of current and future research in vulnerability modeling.

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1. Introduction

The prediction of hurricane losses is of compelling interest to the Federal, State and Local governments, the insurance industry, and homeowners. The Federal and State governments are responsible for enacting policy for reducing the vulnerability of infrastructure (e.g., DMA2K—Law 106–390), as well as, in some states, for regulating insurance rates (Klein, 2009). The insurance industry must assess its exposure to hurricane losses to evaluate risk diversification, reinsurance, and mitigation incentives. (Cardona, 2004; White and Haas, 1975). The public is concerned with the integrity of homes and the cost of insurance.

Awareness of the building stock vulnerability emerged in the 1950s when some Florida counties enacted the first building regulations to standardize construction practice. The insurance industry began around the same time to analyze its exposure and price it using econometric methods based on past catastrophe loss information.

Hurricane Andrew in 1992 changed the perception of the vulnerability of the building stock. A nebulous understanding of this vulnerability resulted in a significant under-prediction of losses and underpricing leaving society unprepared to face the consequences. After hurricane Andrew and the Northridge

earthquake, catastrophe models began to emerge as tools for the prediction of insured losses.

Vickery et al. (2009) present a history of wind hazard modeling. This paper focuses on the development, advantages, challenges, and future trends of the methodologies for loss estimation.

2. Beginnings of loss projections

Hurricane Andrew in 1992 set a landmark in the history of the construction and insurance industries in Florida. Andrew caused \$16 billion in insured losses and \$26.5 billions in total losses in 1992 dollars (Blake et al., 2007; Doebling et al., 1994). The aftermath revealed that the losses could have been significantly smaller had builders complied with and inspectors enforced the existing building code (Khan and Suaris, 1994; Governor's Building Codes Study Commission, 1997). Losses could also have been managed better through prevention, mitigation, and risk spreading, if the loss projection method used by the insurance industry had forecasted higher losses (Chernick, 1998).

2.1. Loss assessments before Andrew

This section reviews how the building codes developed in Florida before 1992, to better understand the evolution of loss estimation methods.

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2.1.1. Building codes system before Andrew

By the 1920s, some cities in Florida adopted the “Recommended Minimum Requirements for Masonry Wall Construction” (Concrete, 1926; USDC, 1925), which was based on collective experience and engineering data (Smith, 1961; Ratay, 2009). Later on, in an attempt to standardize construction practices, some cities and counties in Florida adopted building codes. The earliest examples include Clearwater that adopted a draft of the Standard Building Code (SBC) in 1945 (Cox, 1962), Daytona Beach in 1946 (TMJ, 1946), Bradenton and Manatee counties by 1950, (SJ, 1956), and Riviera Beach in Palm Beach county in 1957 (PBP, 1957). Miami-Dade and Broward counties adopted the South Florida Building Code (SFBC) in 1957 and 1961, respectively. The SFBC was one of the most stringent codes in the United States at the time. Florida’s construction boom of the 1970s subsequently led the state authorities to promote a uniform state-wide building standard by enacting Chapter 553 “Building Construction Standards” of the Florida Statutes (F.S.) in 1974, requiring all counties to adopt a code by January 1st, 1975. The selected allowable minimum codes by the statute were the Standard Building Code (SBCCI, 1975), the South Florida Building Code (SFBC, 1988), the One and Two Family Dwelling Code (CABO, 1988) and the EPCOT code (enforced in Walt Disney World and based on the SBC, SFBC and Uniform Building Code) (RCID, 2002).

Nevertheless, the responsibility for the administration and enforcement was left to the discretion of the 400 local jurisdictions including local governments, local school boards and state agencies (Governor’s Building Codes Study Commission, 1997). The State gave the jurisdictions the power to choose any of the four allowed codes and the authority to amend it as long as this resulted in more stringent requirements.

2.1.2. Problems in the building code system

The wind-load provisions – which improved considerably for low-rise buildings in 1982 (Mehta, 2010) – played a critical role for wind damage mitigation in Florida. The two most widely used codes in Florida, were the SFBC – in Miami-Dade and Broward – and the SBC – in the rest of the state. Although the SFBC was the most stringent code in Florida, this was uncorrelated with compliance and enforcement. In addition, some of the code stringency was eroded for almost three decades (Getter, 1992; Fronstin and Holtmann, 1994). Some of the measures which reduced the effectiveness of the code included the allowance of staples instead of nails for roof decking, thinner roofing-felt, 63 mph rated shingles, inadequate wind pressure coefficients (BCETF, 1992), 6d nails at 12 in. o.c. (panel field nailing) and 6 in. o.c. (roof edges) (Cunningham, 1994), and waferboards – which were less-resistant-when-wet – as a substitute for plywood for roof decking (Fisette, 2005). Furthermore, local amendments made it difficult for engineers, architects and contractors to identify the locally-administered codes and their jurisdictions (Shingle, 2007; Barnes et al., 1991). A study by Florida A&M published in 1987 highlighted deficiencies in code compliance and enforcement in Florida (IBS-FAMU, 1987).

Post-Andrew damage surveys revealed wide spread violations to the SFBC, such as the absence of gable bracing, corner columns, and vertical reinforcing, and the use of gypsum board as wall sheathing (Suaris and Khan, 1996; Cook and Soltani, 1994; Khan and Suaris, 1994; Khan, 1993). Serious shortcomings in the compliance and enforcement process were also observed (Barnes and Mitrani, 1994). Furthermore, for six months after Andrew, more than 60% of the rebuilt roofs in Miami-Dade County did not meet the new SFBC provisions (Chiu, 1994).

Problems with the enforcement of building codes did not disappear rapidly; after hurricanes Opal and Erin in 1995 the

re-building process was delayed due to jurisdictional, enforcement and compliance issues related to the building codes. An expeditious and unambiguous system could have facilitated proper compliance and enforcement, and perhaps could have significantly reduced losses (Governor’s Building Codes Study Commission, 1997).

2.1.3. Econometric models before Andrew

Don Friedman developed catastrophe models for hurricane, flooding and earthquake as early as the 1960s, at Travelers Insurance Co. (Friedman, 1984). Likewise, Karen Clark founded AIR (Clark, 1986), which signaled the emergence of the large scale proprietary catastrophe models. However, up to the mid 1990s the loss projection methods employed by most of the insurance industry relied on historical claim data to set hurricane risk premiums and to estimate average annual losses.

The industry started ratemaking for hurricane peril following the Great Appalachian Storm of November 1950 (Chernick, 1998; Smith, 1950). Hurricane coverage was initially based on existing coverage for perils such as fire. When sufficient hurricane loss information became available, the ratemaking process began to rely on available claims data (Hunt, 1962). By the 1980s up to 30 years of past losses were used for the homeowner policy ratemaking.

In the late 1980s (ISO) sought to improve the existing loss prediction method and reduce rate fluctuation by creating the “excess wind procedure”, which averaged the losses of extraordinary events over the longer time period used for ratemaking. Losses were listed by year and broken into wind and non-wind losses. All years with a wind to non-wind loss ratio exceeding 0.25 were labeled as an excess year, and an excess ratio was calculated. The excess ratio was defined as the wind to non-wind ratios higher than 0.25 minus the median of all wind to non-wind ratios. An excess year was one whose wind losses are higher than the industry average losses for the 30 year period. An “excess loss” was calculated for excess years by multiplying the excess ratio by the non-wind losses. “Non-excess losses” are the excess loss minus total loss for a given excess year. Finally the “excess wind factor” equaled the excess ratios averaged over the 30 years period plus one (Bradshaw and Homan, 1992). The excess wind factor was used to calculate the amount of funds allocated to cope with catastrophe losses.

The methodology described above to project losses was not seriously challenged or revised prior to Hurricanes Hugo and Andrew because of the lack of significant hurricane losses in that period. For 1992, the insurers forecasted \$80 millions in hurricane losses, though the actual insured loss actually was \$16 billion. This forecast compromised the insurance and reinsurance industries, and twelve companies went insolvent (Grossi et al., 2005).

The insurance industry admitted after hurricanes Hugo and Andrew that the use of recent past loss data was not sufficient and a more rational approach was needed (Chernick, 1998). A number of actuaries and researchers suggested that the existing techniques needed to be reinforced with the inclusion of meteorological data and building features to represent the loss hazard more realistically. Other studies proposed to explore the use of simulation techniques in the framework of the excess wind procedure (Bradshaw and Homan, 1992). The shortcomings of the method led the ISO in 1996 to replace the excess wind procedure by simulation methods (Chernick, 1998).

2.1.4. Problems with the econometric methods

The econometric forecast for hurricane Andrew failed because it did not consider the dynamic nature of building-stock risk. The building stock risk is determined by the interaction of

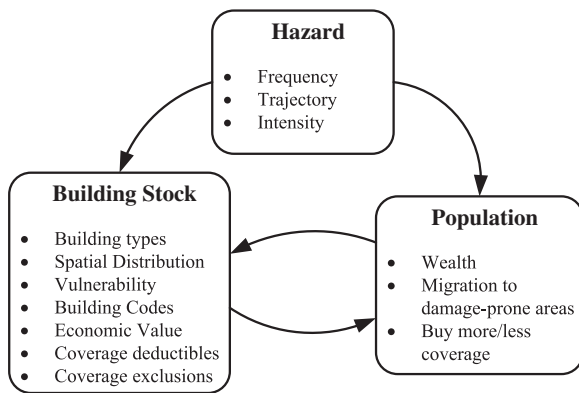


Fig. 1. Diagram of the components of the risk problem and the interactions between them.

time-dependent components such as hazard, population and building stock (Fig. 1). The hazard is described in terms of frequency, intensity and trajectory; the building stock is characterized by real estate market fluctuations, inflation, the building types, their geographic distribution, age deterioration, and building codes enforcement (Friedman, 1984; Davidson et al., 2003); in addition, population trends are characterized by increasing wealth and migration to damage-prone areas (Cutter et al., 2007). In other words, the hazard and building stock have physical, temporal and geographical dimensions, which need to be combined with economic and demographic trends.

The econometric method assumed that the hurricane activity, and consequently the losses, was a steady phenomenon. i.e., the frequency and intensity of hurricanes of the past 30 years were considered a reliable sample for projecting losses.

Regarding the building stock, the econometric method treated the wind and non-wind losses on a uniform statewide basis independently of the geographical distribution of exposure (i.e., relative risk of coastal vs. inland exposure). The possible loss reduction due to more stringent and better enforced building codes was not considered. The heterogeneity of buildings strength, its spatial distribution, and changes in coverage types, limits and deductibles was also ignored. Finally, the method did not consider the increase in losses due to escalating property values and the population migration to the Florida coast (Steinberg, 1997; Pielke and Landsea, 2008; Pielke et al., 1999).

The econometric method reduced the interactive complexity of the above components to a single ratio used to calculate the excess wind factor. The increase in the sample period to account for loss information to predict current and future losses created a false sensation of accuracy. Losses are the result of a unique interaction between a particular hazard and a given building stock vulnerability. As such, past losses are poor indicators of current and future damage (Friedman, 1984). For instance, it is of little value today to know the aggregate damage that a hurricane produced to the Florida building stock in 1950. If the same hurricane hit today, the losses would be different because of the changes in the number of buildings, their value and the current state of existing buildings with aging and/or upgrade to current building codes.

2.2. Post-Andrew Florida response

In order to mitigate losses based on what was learnt from hurricane Andrew, the state of Florida has responded through: (1) the creation of Citizens Property Insurance Corporation to provide property insurance for home-owners who cannot obtain insurance elsewhere; (2) the adoption of the Florida Building Code; (3) the creation of the Florida Commission on Hurricane

Loss Projection Methodology (FCHLPM); and, (4) the development of the Florida Public Hurricane Loss Model (FPHLM). Some these are discussed below.

2.2.1. The Florida building code

After Andrew, concerns arose that a diminution of insurance availability would occur, which would threaten economic growth. In response, Governor Chiles established a Building Codes Study Commission (Governor's Commission hereafter) in 1996 to review the current system of codes. The Governor's Commission found that the existing system had led to a "patchwork of technical and administrative processes" and its recommendations led to the formation in 1998 of the Florida Building Commission responsible for creating a unified Florida Building Code (Governor's Building Codes Study Commission, 1997).

The Commission selected the SBC, developed in Alabama (Ratay, 2009), as the base code for the unified Florida Building Code (FBC), because 64 out of 67 counties were already using the 1973 and the 1997 versions of the SBC with amendments (Shingle, 2007). The SFBC was later included as an additional base code in 1999 to meet the special requirements of South Florida. The Building Commission worked to reach a consensus among stakeholders, and the first version of a unified FBC was made effective on March 1, 2002 (Blair, 2009). Studies have found that the buildings built following the FBC have performed better (ARA, 2002a, 2002b, 2008; Gurley and Masters, 2004). This indicates that the FBC has contributed to reduce the vulnerability of the Florida building stock.

2.2.2. The Florida Commission on Hurricane Loss Methodology

In 1995, the Florida Legislature established the Florida Commission on Hurricane Loss Projection Methodology (FCHLPM) in order to evaluate and certify loss models, and thus create a new ratemaking framework. The creation of the FCHLPM was intended to guarantee "the most actuarially sophisticated guidelines and standards for projection of hurricane losses possible" so that rate filings would not be excessive, inadequate or unfairly discriminatory (F.S. 627.0628(1) (c)). The FCHLPM was formed as a panel of 11 experts that represent the Florida Office of Insurance Regulation (FOIR); the insurance industry, the Florida University System (which provided expertise in finance, statistics, computer design and meteorology), Consumer Advocate, Citizens Corporation, Florida Emergency Management and the Florida Catastrophe Fund (FCHLPM, 2010).

The FCHLPM determined that previous methodologies were not reliable and opened the door to multidisciplinary catastrophe models. The creation of the FCHLPM improved the premium setting negotiation process between the insurance industry and the state and ensured the use of more adequate methods to calculate premiums. The FCHLPM is now the "gold standard for the review of hurricane models for producing property insurance-loss costs" (AAA, 2008) helping all hurricane-prone states by facilitating the development and evaluation of loss models based on standards. For example, Hawaii and Florida only accept models for loss prediction that have been certified by the FCHLPM (AAA, 2008).

3. Catastrophe models

Catastrophe models started a new paradigm for the loss projection methodology. The main historical events and the characteristics of catastrophe models are described and discussed in the coming sections.

3.1. A change of paradigm in loss projection—The advent of catastrophe models

The initial catastrophe models were first developed and applied at the Travelers Insurance Co. by Don Friedman from the early 1960s to the late 1980s. The Travelers' models were used for both insurance and public purposes and laid the bases for subsequent models. Other two early hurricane loss models were developed by the Southwest Florida Regional Planning Council (SWFRPC) and the Texas A&M University. The SWFRPC followed the methodology of Friedman (1975) to project losses (SWFRPC, 1982). The Texas A&M model was developed by Ruch, Berke and Larsen for use in Texas (Ruch, 1984; Berke et al., 1984). This model used Friedman's vulnerabilities. However, catastrophe models, as multidisciplinary simulation-frameworks, emerged and gained wide acceptance as an alternative to the econometric models only after hurricane Andrew (Grossi and Kunreuther, 2006).

New models were developed to consider the aspects of the loss projection problem under a whole new and more holistic perspective. The development of catastrophe models that account for the physics of hurricanes, structural characteristics of buildings and actuarial principles, led to a change in the paradigm of the loss projection strategy (AAA, 2008).

While econometric models are based on "archival-statistical knowledge", catastrophe models are based on "enactment-based knowledge" (Collier, 2008). Both approaches represent different alternatives for assessing risk as described in Table 1. Collier (2008) furthermore argues that catastrophes pushed the limits of the archival-statistical techniques while at the same time enactment techniques emerged and became significant new forms of knowledge and practice.

Catastrophe models – proprietary and public – are now commonly accepted as an improvement to econometric models (Watson and Johnson, 2004). While the private models started in the mid-1990s, HAZUS, the first public model, was launched in 1997 (FEMA-HAZUS, 2006; Schneider and Schauer, 2006). In the early 2000s the state of Florida initiated the creation of the other public model, the Florida Public Hurricane Loss Model (FPHLM) (HB 2145, Ch. 2000-166 (2226); Pinelli et al., 2008).

3.2. Vulnerability estimation methods

The estimates of buildings vulnerabilities are at the heart of a catastrophe model. The vulnerability of a building or its components is usually represented by a curve with a mean damage measurement in the ordinate and a hazard measurement in the abscissa. The building damage measurement might be a ratio of total cost of damage divided by the actual building value. The hazard metric is typically wind speed at 10 m, adjusted for terrain roughness and averaged over a period of time, which is usually 3-s gust, 1 min sustained (in the US) or 10 min sustained (elsewhere). Fragility curves are another way to represent the vulnerability of a building, where the ordinate represents the probability of exceeding a certain threshold of damage for a given wind speed in the abscissa. The concept of vulnerability is used in this paper

in an equivalent manner to the concept of fragility, i.e., as a part of a general framework to assess system performance levels.

Early catastrophe models estimated vulnerability by performing regression analysis on claim data collected by insurance companies. More recent models use engineering principles to project physical damage based on knowledge of wind-structure interaction and component capacities. These engineering models use a probabilistic framework in which the strength of various building components is compared against loading from wind pressure and impact from wind borne debris to assess the likelihood of damage to a given structural type over a series of prescribed wind speeds (Vickery et al., 2003). The physical damage from the engineering model is converted to monetary loss by applying cost estimating and actuarial principles.

3.2.1. Regression vulnerability models

Vulnerability functions derived through regression are intended to describe either the vulnerability of a regional building stock or of an individual building type. The first attempt to relate regional building damage costs to hurricane wind speeds was carried out in 1964 by Don Friedman who fitted a power law model to Florida hurricane data from the previous 78 years (Hendrick and Friedman, 1966; Howard et al., 1972). Ishizaki (1965) reported regional building losses due to the maximum peak gusts of typhoon Vera (Isewan) and Nancy (Muroto II) in Japan. Howard et al. (1972) developed a regional vulnerability curve using coastal damage data from the American Red Cross. Friedman (1984) fitted another vulnerability relationship in 1984 – considered a standard for some time – to Hurricane Donna data. Bhinderwala (1995) studied the correlation between insured damage to wind speed on a hurricane Andrew claim database; the relationship was found to be linear up until interior damage became significant and then highly non-linear. Mitsuta et al. (1996) fitted a power law vulnerability curve to insurance damage and surface wind speed caused by typhoon Mireille in Japan in 1991. Huang et al. (2001) and Sparks (2003) fitted an exponential model to loss data from Andrew and Hugo. Sachs (2007) performed a nonlinear regression analysis to fit a power law model to total damage caused by a hurricane in their entire track.

The regression models described above yielded regional vulnerabilities by considering the building stock as a whole. Other researchers have developed vulnerability models for individual building types. The vulnerability of a region or a portfolio is assessed by aggregating the vulnerabilities of individual buildings in the portfolio.

The first efforts to quantitatively assess the structural damage to individual building types produced by cyclonic winds were pioneered by Leicester and Reardon (1976) in Australia after cyclone Tracy in 1974. Leicester and colleagues also published the first vulnerability curves for individual buildings based on regional damage assessments (Leicester et al., 1979; Leicester, 1981). Their vulnerability curves were proposed as piecewise linear relationships between building damage and 3-s gust wind speeds. Holmes (1996) proposed to replace the piecewise linear model with a lognormal sigmoid model. Stubbs and Perry (1996)

Table 1
Econometric versus catastrophe modeling from Collier (2008).

	Archival-statistical knowledge	Enactment-based knowledge
Data	Archive of past events	Inventory of elements; vulnerability data, event model
Technique of assessment	Statistical analysis of distribution of risks over a population	Enactment of future events

proposed a physically-based assessment of damage coupled with loss data and heuristics. (FEMA, 2000) presented a set of five expert-supplied vulnerability tabular functions for: non-engineered wood, non-engineered masonry, manufactured, lightly-engineered, and fully engineered buildings. Khanduri and Morrow (2003) proposed a methodology to split regional vulnerability curves into individual building types and to translate curves fitted in one region to another. Chock (2005) developed vulnerability curves for many building types in Hawaii based on regression of geo-referenced tax appraisers information, post-hurricane imagery and Hurricane Iniki building losses. The vulnerability relationships between the reconstruction cost (not insurance) and 3-s gust wind speeds were described with a power law model.

Heuristic models were proposed by Mehta et al. (1981), Chiu (1994), Nateghi-a (1996) and Li and Ellingwood (2005). Knowledge-based systems based on fuzzy sets were proposed (Murlidharan et al., 1997, 1999) using damage surveys, questionnaires from experts and data mining (Nawari, 2008).

3.2.2. Assessment: Regression vulnerability models

The method of claim data regression was pursued as a direct path given adequate availability of damage or loss data. Simplicity is the main advantage of regression vulnerability models, but they are affected by the following issues:

1. Most claims records do not include key building features needed to characterize structural types (exterior wall, roof shape, etc.). Consequently, the claims sample would rarely be stratified enough to create a representative vulnerability curve for specific building types. Vulnerability curves thus produced represent a broad aggregate of building types.
2. Catastrophe modelers do not have access to all claims data from any given storm. Some insurance companies sell coverage in preference to certain types of building. Different companies have different claim adjustment practices (Jain et al., 2009). Vulnerability curves derived from these databases could be biased towards a limited portion of the building stock.
3. Insurance valuation issues might affect the usefulness of claim data for assessing building vulnerabilities. For instance, claims above the limits or below the deductibles might be truncated or not recorded, respectively. There could be policies of different types (e.g., actual cash value policies or replacement cost policies). Since the vulnerability curves are meant to represent actual damage before any insurance deductible and limit are applied, the use of claim data to regress vulnerability curves might introduce a bias. The magnitude of the bias is a function of the size of the deductibles and limits, and the mix of insurance policies.
4. Claim databases do not disaggregate losses between building exterior and interior, allowing only the derivation of aggregated loss vulnerability curves.
5. The claim data sample must be large enough to represent the average building type under study to produce unbiased vulnerability curves.
6. Past claim data represent the vulnerability of the past building stock. Andrew claim data provides a depiction of the vulnerability of the 1992 South Florida building stock, which is not representative of current practice. Caution must be exercised if an extrapolation of vulnerabilities through regression in time or space is intended. For the same reason the impact of mitigation measures is difficult to evaluate with regression models.
7. It is difficult to reliably quantify the wind speeds that produced the damage reported in the claim data, adding uncertainty to the models.

8. Claims data do not span all significant wind speeds, particularly those with large return periods.

3.2.3. Engineering vulnerability models

The engineering-based vulnerability methods are designed to predict physical damage based on knowledge of the behavior and resistance of various building components, and their interaction under wind loads. Representations of component capacities are developed based on, for example, test results reported in the literature. Loads are calculated based on building shape and wind speed. Capacity and load are then compared to determine damage for a series of wind speeds. Claims data is not necessary for the development of the vulnerability model, but it is used for validation and calibration. Engineering principles of load and resistance are used rather than exclusively relying on historical observations of failure.

Sciaudone et al. (1997) developed a methodology for assessing vulnerability by combining an engineering approach to model the exterior building damage with data regression to model the interior damage. Applied Research Associates presented a model to assess vulnerability relativities of residential buildings, and the first vulnerability assessment model (by means of relativities) for mid/high-rise buildings model (ARA, 2002a, 2002b). HAZUS is a public model which estimates the building envelope damage by an consists of an engineering-approach and the interior damage by an empirical model based on engineering judgment, insurance claim data and estimations of water ingress (Vickery et al., 2006). The FPHLM (Pinelli et al., 2008) has an engineering model based on Monte Carlo simulations to estimate exterior damage (Weekes et al., 2009) and a model based on engineering judgment to assess interior building damage. The model version for multi-family buildings incorporated a physical account of interior damage based on water intrusion (Pita et al., 2012). HAZUS and FPHLM models are validated with claim data. Sciaudone, et al. (1997) validate against damage surveyed in post-disaster investigations. Other models include Unanwa et al. (2000) who used a fault tree analysis to produce individual vulnerability bands to capture the variability in the damage.

3.2.4. Assessment: Engineering vulnerability models

Engineering models are very flexible, and can be used to derive the vulnerability of a variety of building types and individual features. Advantages of engineering models include the following:

1. Engineering models based on Monte Carlo simulations (e.g., Cope, 2004,) yield probability functions of damage at any wind speed.
2. Vulnerability functions can be derived for a variety of sub-systems, including components, exterior envelope, interior, and utilities.
3. The building types and characteristics that can be modeled are limited only by the existence of reliable information on capacity, and reliable wind–structure interaction models.
4. The effect of mitigation measures and building code improvements can be studied by adjusting the model. For instance, improved nailing schedule for roof decking is modeled by increasing the capacity of that component to uplift damage. Thus the engineering approach is useful for mitigation planning.
5. New engineering data, such as a newly published sheathing capacity study, is easily incorporated within the model. Hence, keeping the model up to the current state of knowledge is straightforward.
6. Engineering models are not subject to the wind speed gaps that claim databases have. That is, the method can project

losses at wind speeds for which no claims data exists. However, the lack of claim data does limit the validation.

One of the disadvantages of these models lies in their complexity. They require more multidisciplinary expertise than regression models. Finally, engineering models regardless of their complexity rely on data for validation. This data might come from insurance claims or damage surveys. The data for validation is the same data used in regression models, and therefore validation is afflicted by some of the same limitations mentioned above.

3.3. Sources of uncertainty in a catastrophe model

Uncertainty in a catastrophe model arises from several of sources for all its model components: wind risk (meteorology), actuarial practice, and building vulnerability (engineering).

The meteorological models are usually statistical simulation models, based on the historical record for the hurricane hazard. Reliable wind data is limited to a period of about 110 years (Powell et al., 2005), and does not reflect the future climate. Terrain roughness values used to convert wind speeds at different exposures are also approximate. More discussion of uncertainties in wind risk models can be found in Vickery et al. (2009) and Powell et al. (2005).

In the actuarial model, the real value of the property might not be identical with the insurance limit recorded in the insurance files. The correlation between building and contents damage is unknown and affects the pro-rated distribution of deductible between the different coverages. There are also variations in claim adjustment practices. More discussion of uncertainties in actuarial models can be found in (Hamid et al., 2010).

Uncertainty within the engineering model arises from many sources. The specific structures that are modeled are generic representations of typical construction, and as such produce an aggregate damage estimate. The translation of wind speeds into wind pressure carries inherent uncertainty in both the wind tunnel methodology (instrumentation, building shapes, approach turbulence, etc.) and the simplifying reduction of fluctuating wind loads into peak scalar values. The damage estimations are influenced by uncertainties regarding the behavior and failure capacity of the various components. Load paths and load sharing are not completely understood for low rise construction, and the models contain assumptions to account for these. Also the interior damage estimation carries uncertainty, especially if the model is heuristic.

The projection of losses in a given region requires as input a probabilistic distribution which is used to represent the existing buildings stock. The data to define this distribution is incomplete, creating uncertainty in the assumed distribution of structure types.

The translation of physical damage to dollar loss (cost of repair) carries uncertainty. Component costs are usually estimated through averages from specialized literature (e.g., R.S. Means, 2010; ENR, 2009, etc.) and expert input. These estimates have a variable margin of error, and can be influenced by temporal changes in material and labor availability as well as claims adjuster practices.

There are numerous contributors to total damage repair cost, and the uncertainty associated with each contributor is different. Fig. 2 is representative of this distribution. It shows the different assembly contributions to the vulnerability as a function of 3-s gust wind speed for a typical old masonry home. It is apparent that interior and utilities are significant contributors to the vulnerability, and that the estimation of damage to these non-structural assemblies (interior, plumbing, mechanical, and electrical utilities) is critical for the accuracy of the model. However,

only limited data exists regarding the vulnerability of building interior and utilities.

The history of code enforcement and the quality of building maintenance cannot be evaluated with great accuracy. The use of year built as a proxy for building strength introduces uncertainty.

The insurance portfolio files contain incomplete information related to the structural characteristics of each policy. The mapping of the available portfolio data into a meaningful input for the vulnerability model is a source of uncertainty. Policies with missing data are handled with vulnerability matrices weighted with factors based on statistical studies of the overall building population (Johnson, 2011; Murphree, 2004).

Modelers are constantly trying to reduce the above-mentioned uncertainties by means of regularly incorporating all new knowledge as it becomes available, and by improving the state of the art of vulnerability modeling through research. The development of hurricane risk modeling is governed by continuous improvements in meteorology, engineering, and actuarial science which aim at reducing these sources of uncertainty. For example: the use of sophisticated instrumentation like satellite tracking, and dropsondes has led to a better understanding of the hurricane phenomenon. The development of new full-scale testing facilities, and the collection of field data during and after hurricane events has led to a better understanding of the action of wind on structures. Research is currently under way to obtain a better characterization of pressure coefficients. Improved data collection by insurance companies results in more complete portfolios. The FCHLPM periodically reviews private and public models and certifies whether they meet its standards. A part of this process is to evaluate the use of the modern knowledge base within the models.

3.4. Challenges to catastrophe models

An important challenge is the user's expectation that catastrophe models should render an accurate estimation of losses. Catastrophe models can provide an estimate of what is likely to happen on an aggregate level over many years and multiple hurricanes, or for a particular scenario. Although the models might be very precise, they are not necessarily highly accurate (Clark, 2009). The uncertainties discussed previously carry over to the estimates of annual expected losses, probable maximum loss, and scenario analyses. Catastrophe models are based on the present incomplete state of knowledge. The challenge for users of these models is to understand the nature of the uncertainties involved, and the fact that the models are meant to provide credible rather than highly accurate predictions of losses.

Another challenge to catastrophe models is their calibration and validation. Since hurricanes are relatively rare events, the loss data available are scarce, typically proprietary, and require filtering to remove biases. Price variations due to fluctuations in the availability of materials and labor (i.e., demand surge) can distort the model outputs significantly if not properly accounted for (Hallegatte et al., 2008).

3.5. Proprietary catastrophe models

Modern proprietary catastrophe models appeared when private modeling firms developed software to project losses in the mid 1980s (Clark, 1986). The first companies in the United States to build catastrophe models were Applied Insurance Research (AIR) founded in 1987 by Karen Clark, Risk Management Solutions (RMS) founded in 1989 and EQECAT founded in 1994. These companies and others acquired considerable expertise in assessing catastrophe damages. While the insurance industry was reluctant to adopt catastrophe models at first, they gained wide

acceptance after Hurricane Hugo and the Loma Prieta earthquake in 1989 and hurricane Andrew in 1992 (Grossi and Kunreuther, 2006).

A debate as to whether commercial models were appropriate to use in the regulatory process centers on the proprietary nature of many assumptions within the models (Nordman and Piazza, 1997; Whitehead, 1997). The Florida Legislature addressed the issue by stating that models would not be “admissible and relevant ... unless the department has access to all factors and assumptions used in developing the standards or models” (F.S. 627.0628). To this end, a Professional Team (henceforth, Pro-Team) was created in 1996 to assist the FCHLPM. It consists of 5 experts in the fields of civil engineering, statistics, meteorology, actuarial and computer sciences with open access to review the rationale, inputs, outputs and assumptions of the model being evaluated for certification by the FCHLPM. The Pro-Team members are not allowed to share its contents publicly. The Pro-Team has published sensitivity and uncertainty analyses of the models (Iman et al., 2002a, 2002b, 2005a, 2005b). Table 2 shows the proprietary models that were certified at least once by the FCHLPM.

The issue of confidentiality in proprietary models has not been entirely settled despite the Pro-Team access to the model details. Regulators perceived wide variations between models outputs, which reduced their willingness to transfer the responsibility of rate filing to proprietary models (Nordman and Piazza, 1997). The “black box” nature of proprietary models renders public scrutiny difficult (Musulin, 1997; Simpson, 2006).

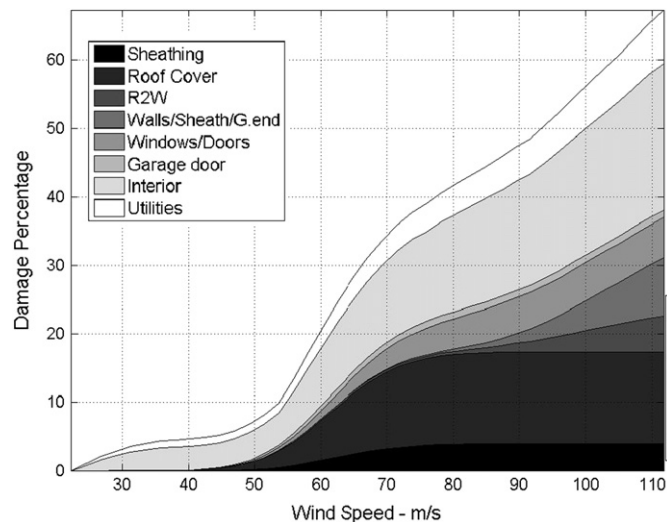


Fig. 2. Components of the vulnerability for a weak masonry structure. (From Pinelli et al., 2008).

Table 2

Proprietary models certified at least once by the FCHLPM. Some are discontinued.

Company	Program	Years certified
AIR-Worldwide	Atlantic Tropical Cyclone Model Classic/2	1996 to 2012
Applied Research Associates	HurrLoss	1999 to 2012
EQECAT	USWIND/ WORLDCAEnterprise	1997 to 2012
E.W. Blanch	Catalyst	1998 to 2000
Risk Management Solutions	RiskLink	1997 to 2012
Tillinghast-Towers Perrin	TopCat	1998

3.6. Public loss models

A public model – though not necessarily open-source – herein refers to a non-proprietary model open to public scrutiny. Examples of public models include HAZUS (Schneider and Schauer, 2006) and the FPHLM (Pinelli et al., 2008). Similar models are being developed in Australia and China (Wehner et al., 2010; Fang, 2009, 2010). In the case of the FPHLM, the Pro-Team and FCHLPM are not prohibited by non-disclosure agreement from publicly discussing the model.

The benefits of a public loss model are manifold: (1) it helps the insurance industry and regulatory authorities in the ratemaking process, (2) it is a research tool to estimate and enhance loss prediction, (3) it is useful to assess regional vulnerabilities and the cost effectiveness of various mitigation strategies, (4) model assumptions are made available to the scientific community and, (5) the process of developing, updating and maintaining the model is conducted in an academic environment. This provides education and training to the students who may become professionals in the hazard modeling communities. The open nature of the public model can be a catalyst for advancing the state-of-the-art, and the exploration of new avenues of research.

4. Future trends

Models to evaluate the vulnerability of the building envelope have become sophisticated. Researchers are now turning their attention to improve the modeling of the interior and contents damage of buildings, which are mainly produced by hurricane-induced rain penetration (Multi-hazard Loss Estimation Methodology Hurricane Model HAZUS[®]MH Technical Manual, 2003; Pita et al., 2012; Pita, 2012; Dao and van de Lindt, 2010). These newest models combine the latest meteorological research on rain during hurricanes, including rate, duration, drop size, impinging characteristics (Straube, 2010), and tests results of water intrusion and propagation through the building envelope (Salzano et al., 2010; Masters et al., 2010). Although vulnerability models already represent damage caused by the joint action of wind and rain intrusion, more research is needed. In particular, the wind engineering community is looking at possible new weather “intensity” parameters to capture the correlation between wind and rain damage.

Mid and high-rise buildings exhibit a large variation in shape and size, and therefore are not amenable to be classified into a few manageable generic models. The vulnerability of these structures is being explored through new techniques based on a modular approach (Pita et al., 2009, 2012; Pita, 2012). In this case, the vulnerability of complex buildings of any shape or size can be modeled as the aggregation of the vulnerability of modules (e.g., apartment or office units).

An important aspect of a building vulnerability is also related to loss of use, even in the event of no direct damage to the building. Buildings are by their very nature integrated into their surrounding local, regional, or even country wide (and beyond) infrastructure. This infrastructure includes networks of transportation, power, water distribution, and communication (Reed et al., 2009; McDaniels et al., 2007; Reed, 1999). Future vulnerability models will integrate the vulnerability of residential and industrial buildings with the vulnerability of infrastructure networks. The result will be a more accurate prediction of economic losses due to hurricanes, including time related losses, and their macro economic consequences (Tatano et al., 2001).

Geographic information systems (GIS) will play an increasing role in vulnerability assessments. Layering input data and vulnerability model results will increase the capacity for data analysis and processing and improve emergency management and planning.

Catastrophe models and regional risk analysis rely on accurate quantifications of the building portfolio. It is common that the building databases for risk assessments i.e., insurance portfolio files, claim databases, tax appraisers databases, etc., lack information on important building features. Procedures to infer the missing information in a database from the available information are being explored with probabilistic methodologies (Pita et al., 2011b).

Finally, an increased international collaboration for open-source vulnerability methodologies is currently underway with organizations such as Alliance for Open Global Risk Analysis (AGORA), and the Global Earthquake Model in the field of seismic engineering (<http://www.globalquakemodel.org>).

5. Summary

This paper addresses the development, advantages, challenges, and future trends in the techniques to assess building vulnerability in loss projection models.

A multidisciplinary simulation-framework perspective replaced the econometric models after hurricane Andrew. The new catastrophe models considered many key variables of the loss projection problem, their interactions, and the uncertainties involved.

Two main approaches have been developed for assessing building vulnerability, namely claim regression and engineering vulnerability. Regression of claim data only captures the vulnerability of past building stock. Although the model is straightforward, the claim data usually does not convey structural type information. Engineering models, on the other hand, do not rely on past historical observations of failure but apply physical principles of load and resistance. The challenges to engineering vulnerability models lie in their complexity, and in the validation and calibration process.

Catastrophe model components (vulnerability, actuarial and meteorological models) all carry epistemic uncertainty. The potential exists to reduce uncertainty as improvements in the state-of-the-art of the disciplines evolve.

Current and future research is focusing on better estimation of interior and contents damage, as well as time related expenses. The vulnerability of portfolios of more complex structures, such as mid/high-rise buildings, is being investigated. There are also efforts to model the interdependence of building and infrastructure vulnerability to provide complete pictures of damage, including long term economic consequences and estimates of recovery cost and duration. All these new lines of research respond to an increasing trend of international collaboration for catastrophe research.

Acknowledgments

This research is supported by the State of Florida through its Office of Insurance Regulation grant to the International Hurricane Research Center, at Florida International University. The opinions, findings, and conclusions expressed in this presentation are not necessarily those of the OIR. The research was also funded by the Center of Excellence for Hurricane Damage Mitigation & Product Development at Florida International University. The support and advice of Prof. Judy Mitrani-Reiser at John Hopkins University is also gratefully acknowledged.

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